

1. Project Planning

Our team chose to use a method to do the Stage 1 design part of the Build Safe Synthetic Test Environment for MedTalk Connect Interoperability for FAZ Australias project. We used Jira as our tool to manage the project. We worked in two areas: HEAL for weekly results that track how the industry's project is doing and HITH for breaking down tasks for tracking history of Assignment 1. Also, GitHub helped keep all the documentation in one place. Microsoft Teams helped us talk every day and have weekly check-ins with our Industry Supervisor, Atif Nisar.

My own challenge started from the beginning when I was not sure what tasks I would get. At first I thought my job would include the project charter (Week 1) the full Software Requirements Specification (Week 4) and all the architecture diagrams. After a team meeting my work was made clear. I was now responsible for Week 5. Feasibility and Risk Analysis (HITH-6). This work was supported by my deep-dive documents: design-error-handling.md, failure-mode-test-plan.md data-validation-rules.md and standards-review.md. To show this change I changed my Jira board to show the eight tasks I had (HITH-21 through HITH-29). I started by making the designs first before working on the feasibility report. This way every part of Week 5 had proof of just guesses. Because of this I learned how important it is to plan from the bottom up. Making sure there is evidence, before putting big ideas.

2. Teamwork

I was part of a 4-person team. I performed the function of risk and feasibility lead as part of the team and as my responsibilities included performing test cases to see how the system dealt with mal formed HL7 segments, dropped TCP connections and cases where a mock connector is unavailable. I designed a system for tenant isolation with row level access and a token-based algorithm to implement rate limiting.

I wrote the feasibility report document in Week Five and the associated technical, operational, legal and economic sections. The team communicates well however it required boundary setting at times, primarily when my Week Two Jira tickets which overlap with the team's week one, in Weeks One and Four, were required to be done and were able to align these with another team member's tasks through a shared scope document and work against the HITH requirements through use of a RACI matrix. Working within this environment combined clinical practice and domain knowledge (FHIR, HL7, SNOMED CT-AU) and software engineering I found the differences in thinking regarding safety, for clinicians compared to developers was large.

I learnt the importance of bridging between clinical and technical domain languages equal to the importance of writing code itself, and the need for clear and concise documentation for team members with differing technical depth e.g.(README.MD, commits and Jira tickets).

3. Integration of Learning

This project required a deep combined understanding of nearly every unit that I have learnt so far. Cybersecurity taught me the concept of defence-in-depth, which I used during security implementation, rate limiting features and isolate tenants. That is to say, I was able to make sure that any security lapse would not have a significant impact on my synthetic dataset. I further used my networking know-how in establishing simulated MLLP connections and TCP resets through which I came to grips with how sockets work and configured transport-level transport timeouts accurately.

Database Systems gave me an advantage to implement data validation whereby fake IHI's were generated and referential integrity was ensured for FHIR and SNOMED CT-AU. In Software Engineering, I made judicious use of software engineering theories to help in writing design operation outcomes and requirements traceability matrices and to separate out my test code from fault description files. Cloud computing gave me grounds to use Docker. With Risk Management, I had enough insight into drafting my risk log which involved an analysis of risks in the probability x, impact matrix and mapping of mitigation controls to fault modes test cases.

Project Management guided me for my feasibility assessment and used Jira for task breakdown, stakeholder management and facilitating communications with Atif Nisar (industry supervisor) for this project. This project is a projection of real world systems integration which cannot be solved in silos, it requires holistic thinking across networking, security, data, and governance.

4. Professional Development

My technical take-aways are: ability to both read and validate FHIR R4 Patient resources; to parse HL7 v2.4 messages and to relate SNOMED CT-AU codes to ICD-10-AM and MBS codes. My personal professional take-aways relate to gaining further skill writing technical design documentation for a professional audience-this wasn't just academic essay writing but generating the kind of specs needed with defined json schemas, http codes and test cases that make for production readiness. On the flip side, scope estimation wasn't my initial forte (I committed to too much, far too much across many weeks, prior to the team clarifying with me the scope), but this has been addressed through much more detailed estimation techniques going forward and requiring written commitments to task delivery before commencing work on a task.

Also, as thorough as my documentation skills are, this project has highlighted an actual coding deficiency – I'll also be developing hands-on coding in Python with FastAPI, and HAPI FHIR servers for a next attempt at stage 2 of such a project; as well as, learning more about Australian healthcare regulations in the Australian health and medical space beyond the privacy act 1988, such as the MY Health Records Act, or individual state legislation to ensure clinical governance frameworks are put in place etc..

The experience I've had has been invaluable, turning into my first real exposure of what it takes to create a safe, Standards compliant application within regulated environments. Through using Synthetic Data, Isolation of tenants as well as injecting failure into systems, this exposure has taught me not only the technical terminology as well as how to work along with software engineering as well as digital health.